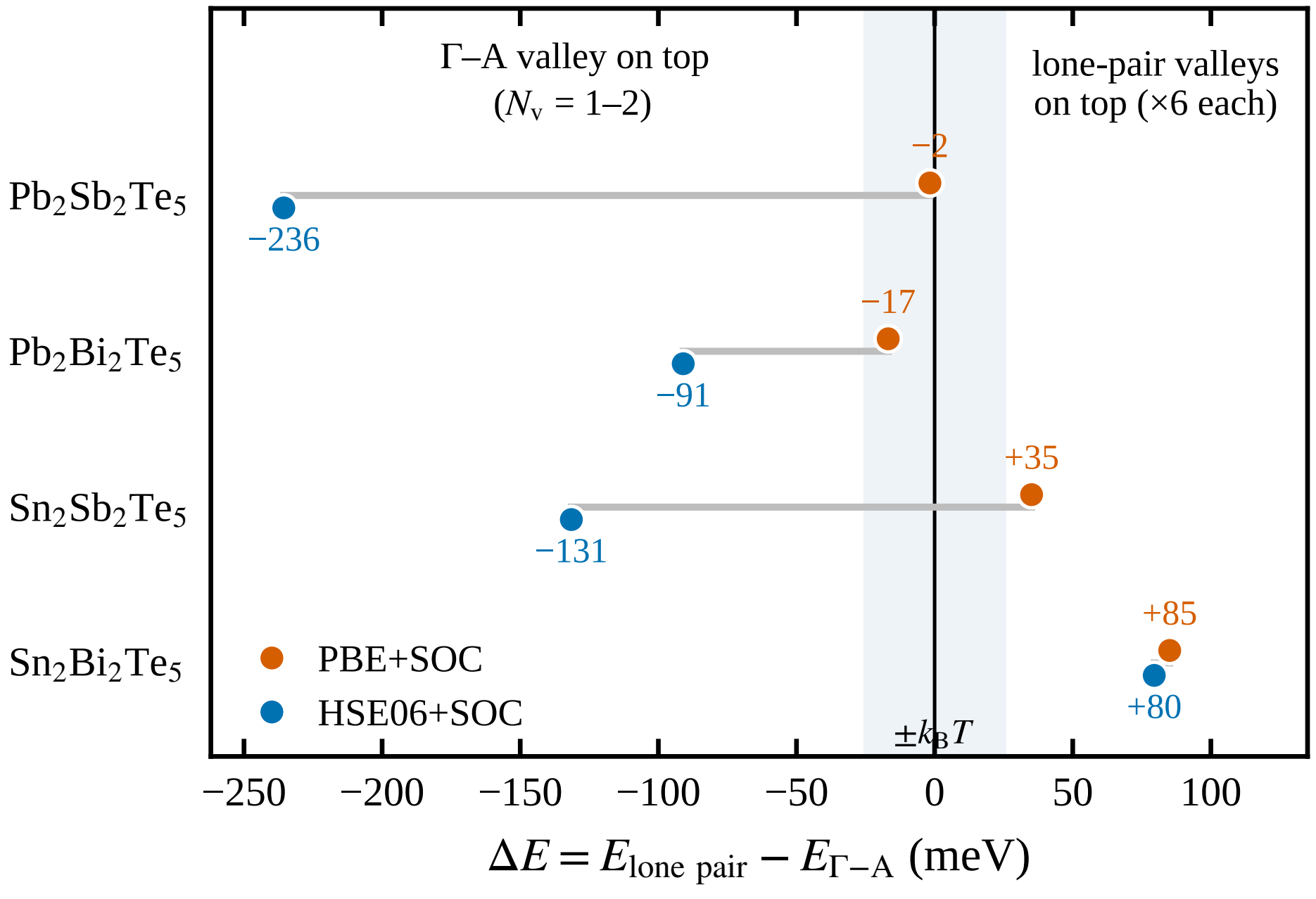
Γ–A valley on top
(N_v = 1–2)
lone-pair valleys
on top (×6 each)
$Pb_2Sb_2Te_5$
$Pb_2Bi_2Te_5$
$Sn_2Sb_2Te_5$
$Sn_2Bi_2Te_5$
−2
−236
−17
−91
+35
−131
+85
+80
PBE+SOC
HSE06+SOC
$\pm k_B T$
−250
−200
−150
−100
−50
0
50
100
$\Delta E = E_{\text{lone pair}} - E_{\Gamma-A}$ (meV)